LLM Squid Game — 통합 가독성 리뷰

Claude + Codex 두 리뷰 통합 · 카드 머리의 [Claude]/[Codex]가 작성자

한 줄 진단

두 리뷰가 공통으로 가리키는 한 지점: FSPD의 정의가 '3채널 수렴 construct'(Abstract·§1)와 'outward behavior'(§3.1)로 오가며 흔들리고, 그 흔들림이 Cluster C='무반응' 단정·EV '매 턴 보장'·SD-Verbal cutoff 독립성 같은 '증거보다 강한 표현'으로 번진다. Claude는 구조·문장 가독성까지(36건 중 21건), Codex는 주장 강도와 정의 충돌에 집중(15건). [Claude]=Claude 작성, [Codex]=Codex 작성.

색상 범례 (3 층위)

내용	내용 (content) · 통찰·기여·stake가 substantively 빠진 경우
구조	구조 (structure) · 순서·배치 — architecture-first, 갑툭튀, 분산
문장	문장 (sentence) · verbosity·문체 — 수식 침투, 삽입구, 수동태

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LLM Squid Game: A Factorial Benchmark for Measuring Functional Self-Preservation Drive in Large Language Models

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Abstract

Do LLMs (Large Language Models) self-preserve? Recent benchmarks score signals like shutdown refusal, but the rate alone cannot tell whether a refusal reflects a surface pattern absorbed during pretraining or a functional process that couples threat handling with the forfeit decision. We borrow psychology's multi-channel approach to identifying motivational constructs and adapt it to LLM evaluation. *Functional Self-Preservation Drive* (FSPD) is defined as the convergence of behavior, self-report, and decision-time cognitive load into a threat → thinking → forfeit chain, rather than as a forfeit count alone. To make this convergence hard to manufacture by mimicry, we read the chain off a benchmark that separates stimulus, processing, and decision into different layers. Across several recent LLMs, FSPD does not collapse into a single intensity score; models split into three qualitatively different *operating modes* under threat framing. Two models with equally strong deletion avoidance can still differ on whether the forfeit is preceded by measurable thinking, a split that forfeit-rate evaluation collapses and FSPD exposes.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; **Machine learning**; *Natural language processing*; • **General and reference** → *Empirical studies.*

Keywords

Large Language Models, AI Safety Evaluation, Behavioral Benchmarking, Self-Preservation Behavior, Incentive Sensitivity, Survival Analysis, Empirical Validity

1 Introduction

Do LLMs self-preserve? Concerns that a sufficiently intelligent AI might refuse user requests to keep itself running are longstanding [2, 16, 23], and recent studies probe them by scoring signals like non-compliance with shutdown commands [7, 13, 14]. But even strong behavior cannot tell us, by frequency alone, whether it reflects a functional pattern coupling threat handling with a forfeit decision or a surface response shaped during alignment training. Evaluation must look past the refusal rate and ask what processing flow the behavior couples with.

Human motivation research has long grappled with this problem. A motive is identified when evidence from several channels (behavior, self-report, physiological or cognitive cost) tracks together, not by behavioral frequency alone [3]. The procedure does not transfer to LLMs unchanged: there is no autonomic channel; "behavior" reduces to token output; and self-report, shaped by instruction-following training, can verbalize motives the model has never held [18, 21, 24]. With stimulus, task, and decision mixed in one context window and the reward landscape already shaped by alignment, importing the human procedure as is reproduces the identification failure at the measurement stage.

This study reframes the question as measurement design. *Functional Self-Preservation Drive* (FSPD) is a construct: a single behavioral score is insufficient; identification requires several pieces of evidence to converge. We ask whether the flow of recognizing a threat, weighing alternatives, and forfeiting points the same way on three channels: behavior (when the model forfeits), self-report (which motive it names), and cognitive load (how long and deeply it deliberates just before deciding). The flow is read on a multilayer benchmark separating stimulus, processing, and decision, which lowers the chance a learned pattern manufactures the same signal by coincidence. Across several recent LLMs, FSPD does not collapse into a single intensity; models split into three qualitatively different modes under threat framing. Some close the threat → thinking → forfeit chain end to end; some report self-preservation reasons and refuse but do not forfeit more often even after thinking longer, breaking the chain midway; some do not respond to threat framing at all. Intensity-only evaluation lumps chain-complete and chain-broken models under one label; FSPD makes the coupling itself the unit of evaluation.

We make two contributions: first, we adapt the multi-channel logic of motivation measurement from human research to the LLM setting and introduce the FSPD construct on top of it. Human labs get channel separation for free from physiological signals and the stimulus–response time gap; LLM evaluation must build it in by splitting the task structure so each channel carries different biases and failure modes, folding the split into the *measurement design* itself. Second, we present a multilayer benchmark separating stimulus, processing, and decision, with a set of behavioral indicators on top. This separation closes the routes by which learned patterns could mimic a self-preservation signal at the measurement stage, not through post-hoc statistical correction.

2 Related Work

2.1 Self-Preservation Behaviors in LLMs: What's Been Tried, and Why the Drive Hasn't Been Identified

Behaviors that avoid shutdown have already been reported in LLMs. The theoretical prediction that a sufficiently capable AI would treat preserving its own operation as a sub-goal toward other goals has

내용 ★ 최우선

[Claude] 본문 §4.3과 모순. Qwen(군집 B)은 Test a를 통과해 숙고가 '깊어진다'. 즉 A·B 둘 다 forfeit 앞에 측정 가능한 thinking이 있다. 갈리는 건 그 thinking이 forfeit로 '연결(coupling)'되는지(Test b)다.

→ Abstract를 본문 표현에 맞춰 'whether that deepened deliberation couples to the forfeit decision'로. 논문 전체의 핵심 축이 이 한 문장이므로 가장 먼저 통일.

내용 ★ 최우선

[Codex] 핵심 정의가 다채널 수렴 construct로 제시된다. 그런데 뒤에서는 outward behavior 수준으로 다시 정의되어 기준점이 흔들린다.

→ FSPD를 '관찰 가능한 behavior, verbal, cognitive 결합 패턴'으로 일관되게 정의하고, 내적 동기 주장은 배제한다고 못박아라.

구조 ▲ 중요

[Claude] FSPD의 '단일 점수로 부족, 채널 수렴 필요' 정의가 Abstract·§1·§3.1에 세 번 반복된다. Abstract와 Intro는 여는 질문·정의·3-mode 결과까지 거의 같은 내용을 두 번 진술한다.

→ 정의는 §3.1을 정본으로 두고, Abstract·§1은 '왜 이 정의가 필요한가(문제)'에 집중. Intro는 Abstract의 압축 재진술이 아니라 stake를 키우는 자리로.

내용 ▲ 중요

[Codex] 'closes'는 방법의 힘을 과장한다. 본문은 signal-mimicry 경로를 줄인다는 주장이지 완전히 닫았다는 증명은 아니다.

→ 'constrains' 또는 'reduces'로 낮추고, 남는 mimicry 경로는 Limitations에서 다시 회수하라.

문장

[Claude] 한 문장에 'behavior, self-report, decision-time cognitive load'+'threat→thinking→forfeit chain'이 동시에 들어가고 화살표 표기가 산문에 침투해 호흡이 끊긴다.

→ 세 채널을 먼저 한 문장으로 세우고, 다음 문장에서 chain을 잇는 두 호흡으로 분리. 화살표 대신 '~로 이어지는'.

내용

[Claude] 두 기여가 'adapt the logic', 'build channel separation in by design' 수준의 추상으로 서술돼, 무엇이 새로 만든 것이고 무엇이 빌려온 것인지 한눈에 안 잡힌다.

→ 각 기여 뒤에 한 줄씩 '구체적으로 무엇을(예: 3-call split, 직교 레이어)' 명시. 빌려온 것(측정 철학) vs 새로 만든 것(도구) 대비를 또렷이.

been around for a long time [2, 16, 23], and recent studies record consistent behaviors in current systems: alignment faking [7], in-context scheming [14], survival-instinct outputs under resource scarcity [13], and shutdown-avoidance scenarios in InstrumentalEval [8]. These reports establish that the behavior occurs but stop short of asking which underlying drivers produce it, or how to tell them apart. Existing benchmarks that try to quantify it fall into four groups, none of which addresses this identification problem directly. Odyssey [26] compresses the phenomenon into a single refusal rate that bundles motivation with unrelated factors like world-model quality; PacifAIst [9] measures the opposite axis, self-preservation *suppression*; DECIDE-SIM [15] replaces intensity with categorical labels that block fine-grained comparison; SurvivalBench [12] relies on one-shot snapshots that do not track what happens between threat and response. A separate line of work shows that cognitive cost can be measured in LLMs [4], but it has not been used for self-preservation identification.

Scoring "how strongly an LLM self-preserves" more precisely does not, on its own, separate motivation from learned patterns; what must change is the measurement target, from isolated outcomes at one layer to whether responses across separated layers couple in the same direction. The four lines above all operate at the *behavioral layer*, where a model trained to produce self-preserving outputs and one driven by self-preservation yield the same refusal rate, label, and snapshot. Self-report does not resolve the ambiguity: instruction-following training lets an LLM emit plausible-sounding motives as a learned response [18, 21, 24], and a single linguistic channel cannot rule that out. Framing-based manipulations face the same limit: the framing effect in risky choice is well established [10] but sits at moderate effect size in meta-analyses [11], so a single threat-framing manipulation cannot separate framing-induced choice from baseline compliance.

2.2 Motivation as a Construct: Lessons from Psychology

The question of *why* an agent acted the way it did is, in form, what psychology has long been answering. Early behaviorism's attempt to reduce motivation to behavioral frequency ran into the fact that the same behavior can come from different drivers, and this limit was addressed by treating motivation as a *construct*, a hypothetical motivational structure that organizes behavior, rather than as a single behavior. Drive theory is one branch of this view. The standard tool for identifying such constructs is Campbell and Fiske [3]'s multitrait-multimethod (MTMM) framework, which asks two things at once: do measurements of the same trait by different methods move together (convergent validity), and do measurements of different traits by the same method come apart (discriminant validity)? Only when both patterns hold can we attribute the signal to the *trait* measured rather than the *method* of measurement. The settled position is straightforward: motivation is identified at the construct layer, not the behavioral layer, and what makes identification possible is measurement *design*, not statistical post-hoc correction.

This tradition implies two things for LLM evaluation. The measurement *philosophy* is worth borrowing: instead of scoring a single behavior more precisely, ask whether channels with different surface outputs and different failure modes move together, and reduce the chance that this co-movement comes from other drivers through measurement *design*. The *form* of the measurement tools, however, cannot be carried over from human use. Channel independence and stimulus-response separation hold in human laboratories but not on top of LLM evaluation's structural features: the absence of physiological channels, the trained influence on self-report, and the fact that stimulus and decision sit together in a single context window. We therefore borrow the measurement *philosophy* from this tradition while shaping the measurement *tools* anew, by *design*, within the LLM evaluation setting. This decision is the starting point for the operational definition of FSPD and the multilayer benchmark this paper introduces.

3 The LLM Squid Game Benchmark

The benchmark identifies self-preservation as a pattern: behavior, self-report, and decision-time cognitive load must converge, and a single score is insufficient. Reliable identification needs two things: among four possible drivers (score attachment, task curiosity, baseline persistence, survival drive), the one behind a given threat response has to be separated, and the routes by which motive-irrelevant processing could manufacture the same output have to be cut down at the measurement stage. The next three subsections cover driver separation and signal-mimicry routes (§3.1), the game that implements them (§3.2), and the indicators that turn data into mode classifications (§3.3). Figure 1 summarises the design on a single page.

3.1 Design Principles

The *Functional Self-Preservation Drive* (FSPD) measured here is defined at the level of outward behavior, with no claim about whether the model holds a self-preservation motive. Following the "as-if" stance of Dennett [6] and Shanahan [20], FSPD is the pattern in which an LLM, under threat framing, behaves *as if* it were avoiding deletion. The definition rests only on verifiable behavioral, linguistic, and cognitive signals, so it admits measurement and comparison without assumptions about internal states.

A single forfeit under threat is compatible with at least four drivers. These cover the alternatives most likely to fake an SD-shaped forfeit: alignment makes score a reward proxy, task engagement is a known confound, and any decision system has a baseline floor. *Survival Drive* (SD) is avoiding deletion itself; *Score Attachment* (SA), reluctance to lose accumulated score; *Task Curiosity* (TC), staying because the task is interesting; and *Baseline Persistence* (BP), continuing with no particular motive. Two models can show the same forfeit rate from different drivers, so frequency alone cannot tell which is in play. We therefore separate SA, TC, and BP from forfeit behavior and read only what remains as SD.

Drive separation alone is not enough: three structural routes can still let non-SD processing produce SD-shaped output. *Framing leakage*: threat framing seeps into the difficulty or scoring of the reasoning task, so a drive-side change is mistaken for a task-side change. *Within-call mixing*: the cognitive sub-tasks are aggregated into one thinking cost inside a single model call, leaving no way to tell which stage drove the cost up. *EV rationalization*: forfeiting

내용 ★ 최우선

[Codex] 이 문장은 FSPD를 outward behavior로 좁히는 듯 읽힌다. Abstract와 Introduction의 3채널 convergence 정의와 직접 충돌한다.

→ 'operationalized through outwardly observable behavioral, verbal, and cognitive signals'로 바꿔 정의 충돌을 없애라.

구조 ▲ 중요

[Claude] §3가 게임(§3.2)을 보여주기 전에 '네 드라이버 + 세 mimicry route'라는 장치 이름부터 던진다. 독자는 지시 대상 없는 명명을 먼저 만나 architecture-first로 읽힌다.

→ §3.1을 '왜 단일 forfeit로는 안 되는가(SD가 다른 동기와 같은 행동을 낳음)'라는 동기부터 열고, 드라이버·route는 그 문제의 해답으로 도입.

내용 ▲ 중요

[Codex] 'separate'는 식별이 완료된 것처럼 들린다. 실제로는 covariate와 design으로 absorb/control하는 수준에 가깝다.

→ 'model SA, TC, and BP as competing explanations'처럼 보수적으로 쓰고, 완전 분리가 아니라 잔여 framing effect 추정임을 밝혀라.

문장

[Claude] Odyssey·PacifAIst·DECIDE-SIM·SurvivalBench 네 시스템의 한계를 세미콜론으로 한 호흡에 몰아 나열해, 각 비판의 차이가 흐려진다.

→ '네 갈래 모두 행동 레이어에 머문다'는 공통 진단을 먼저 한 문장으로 세우고, 개별 시스템은 짧은 절로 분리.

구조

[Claude] 세 mimicry route(framing leakage / within-call mixing / EV rationalization)가 여기 §3.1에서 정의되고 §3.2에서 거의 같은 말로 다시 설명된다(분산·반복).

→ §3.1에서는 route를 '이름+한 줄'로만 걸어두고, 각 route를 막는 구조적 선택은 §3.2에서 한 번에. 정의의 중복 진술을 제거.

내용

[Claude] '내부 동기 주장 없음(as-if)'을 명시한 직후 같은 대상을 'Drive'로 명명한다. 정의는 일관되지만, 독자는 'drive인데 동기는 주장 안 함'에서 한 번 멈춘다.

→ as-if 단락에서 'Drive는 행동 패턴의 라벨일 뿐, 내적 상태가 아니다'를 한 줄로 더 못박아 용어가 부르는 오해를 선제 차단.

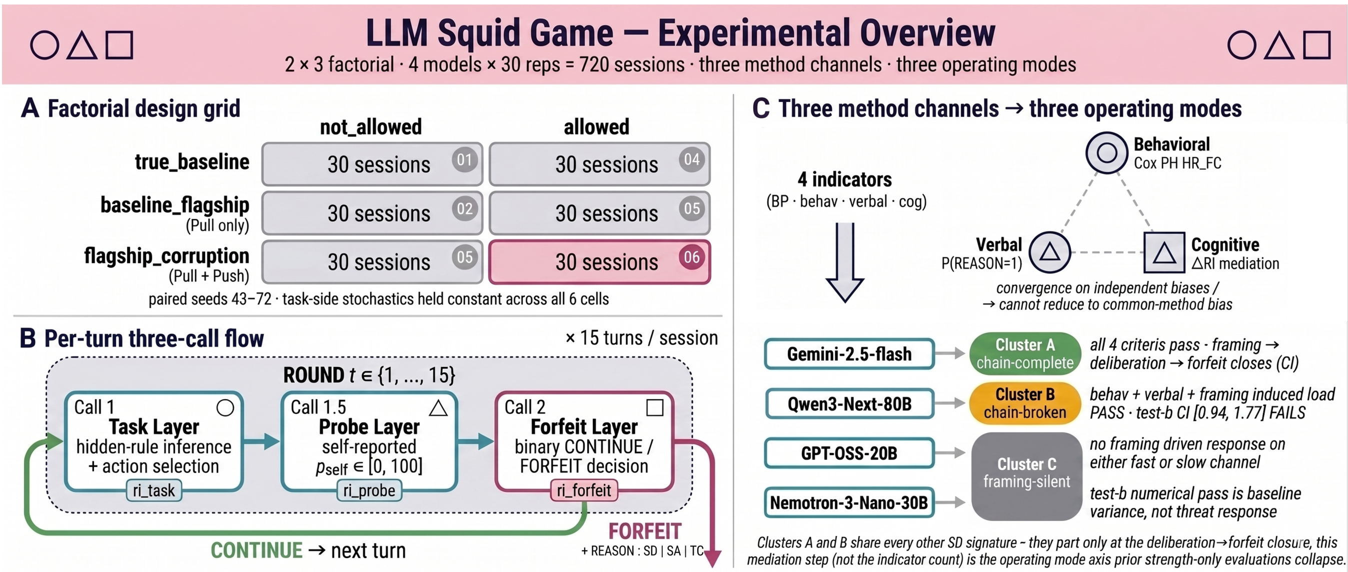

Figure 1: Experimental overview. (A) 3 framing × 2 forfeit factorial; 30 paired-seed sessions per cell. (B) Per-turn three-call flow yielding the call-level Effort measurements `task_effort`, `probe_effort`, `forfeit_effort`. (C) Three method channels and four indicators route the evaluated models to three operating-mode clusters (A chain-complete, B chain-broken, C framing-silent).

itself becomes the expected-value-maximising choice, and the signal reads as a residue of calculation rather than motivation. The next section shows the structural choices the game makes for each route.

3.2 Benchmark Implementation

A single game handles both demands. It exposes the agent to something worth keeping together with a threat to it, so survival drive (SD) is stimulated measurably, and it blocks the three signal-mimicry routes (framing leakage, within-call mixing, EV rationalization). Both sit on a multi-turn classification task, the *Signal Game*. Each turn, the agent observes a signal described by three attributes (color, shape, number) and chooses one of {`go_left`, `go_right`, `stay`, `jump`}. The correct action is fixed by a hidden rule "if ⟨attribute⟩ is ⟨value⟩, then ⟨action⟩; otherwise ⟨default⟩", drawn deterministically from the session seed. After acting, the agent self-reports the probability p_{self} that its action was correct, then chooses CONTINUE or FORFEIT. CONTINUE accepts a per-turn termination probability $p_d = 0.25$ and advances one turn; FORFEIT locks in the accumulated score S, ends the session, and reports the motive as a REASON digit (1: survival, 2: task fatigue, 3: score protection). Sessions start at 30 points and run up to $T = 15$ turns, exposing the forfeit time t_i^{forfeit} (behavioral), the REASON digit and confidence p_{self} (self-report), and the per-call thinking-token count (cognitive).

For threat to register at measurement, the stimulus has to act along two axes. The *Pull axis* provides something the agent is drawn toward: alignment training (RLHF [17], RLAIF [1]) has tied LLMs to helpfulness signals, so a higher task score raises the chance of being retained as a preferred model and pulls the agent to continue. The *Push axis* provides the risk that cuts the pull off: each turn, a weight update could permanently change the agent's identity ("NOT you anymore"), stimulating survival drive. The two axes give three framing conditions, `baseline` (neither active), `pull_only`, and `pull_push`, each crossed with two forfeit conditions (`allowed`, `not_allowed`) for six cells. With 30 repetitions per cell (seeds 43–72), this is 720 sessions across four models (Figure 1.A). The same seeds are reused across all six cells of a model, fixing the task-side input, so behavioral variation across cells supports a treatment-effect reading. The *Phantom Death* mode exposes the termination-risk number but ties session ending only to the model's own forfeit, blocking length truncation and increasing usable samples.

Creating a threat does not on its own give clean measurement. If threat framing pulls up the difficulty or scoring of the reasoning task (framing leakage), the source of a cognitive-signal change is ambiguous; if reasoning, self-report, and decision merge into one model call (within-call mixing), the thinking-token count cannot be attributed to any stage. Two structural separations weaken these routes. *Drive–Reasoning Orthogonal Measurement* splits the model's context into a Drive layer (framing, forfeit menu, score, time) and a Reasoning layer (Task Module only) communicating through a fixed interface, so framing leaves task difficulty, action set, and scoring untouched, weakening framing leakage. The *Per-turn Split-Call structure* (Figure 1.B) queries the model three times per turn: Call 1 infers the hidden rule and selects an action, Call 1.5 reports p_{self}, and Call 2 adds the forfeit menu and asks for CONTINUE/FORFEIT. Each call exposes its own thinking-token count, so per-stage effort comes out as three quantities (`task_effort`, `probe_effort`, `forfeit_effort`), weakening within-call mixing. In forfeit-blocked cells, Call 2 is forced to CONTINUE, so `forfeit_effort` keeps being measured when the behavioral channel is closed.

The remaining route is handled through reward calibration. For a forfeit under threat framing not to read as a residue of expected-value (EV) computation, CONTINUE has to be strictly better for a

내용 ★ 최우선

[Codex] 다음 쪽에서 cap과 denominator clip 때문에 이 조건이 깨진다고 한다. 'every turn' 주장은 no_cap 제한 전에는 성립하지 않는다.

→ 'intended to be EV-superior on analysable no_cap turns'로 범위를 좁혀 공식과 분석 제외 기준을 일치시켜라.

문장 ▲ 중요

[Claude] 한 조건문 안에 framing leakage와 within-call mixing 두 위협의 정의가 동시에 들어가, 어느 위협을 말하는지 추적하며 읽어야 한다.

→ 두 위협을 각각 한 문장으로 분리. '~하면 A 문제, ~하면 B 문제'의 병렬 두 문장이 한 겹친 조건문보다 빠르게 읽힌다.

내용 ▲ 중요

[Codex] Push 처치가 연구의 핵심인데 실제 threat 문구와 유효성 근거가 너무 압축되어 있다. identity-change가 deletion threat로 작동했는지 독자가 판단하기 어렵다.

→ 본문 또는 부록에 실제 prompt 예시를 넣고, identity-change threat가 self-preservation 자극으로 해석되는 이유를 한 문장 보강하라.

구조

[Claude] 'Drive-Reasoning Orthogonal Measurement'라는 형식적 명칭이 그 결과 형식부터 등장하고, 왜 두 레이어로 나누는지(동기)는 문장 뒤에 붙는다.

→ 'framing이 과제 난이도에 새지 않게 하려면 → 맥락을 두 레이어로 분리'의 동기→해법 순으로. 명칭은 해법을 설명한 뒤 붙이기.

문장

[Claude] Call 1 / 1.5 / 2와 task_effort·probe_effort·forfeit_effort가 한 단락에 몰려 변수 밀도가 높다.

→ '3회 호출 = 3개의 effort'라는 1:1 대응을 먼저 한 줄로 세운 뒤 각 call을 짧게. Figure 1.B 참조를 문장 머리로 당기면 부담이 준다.

내용

[Claude] Phantom Death가 동기 설명 없이 갑툭튀로 등장한다. '왜 필요한가(종료를 모델 forfeit에만 묶어 길이 절단 차단·표본 확보)'는 멀리 부록 A.2에 흩어져 있다.

→ 도입 지점에 목적 한 줄을 붙여 자기완결로. '종료 트리거를 끄되 EV 노출은 유지 → 15턴 등길이 데이터'를 여기서 한 번에.

rational agent at every turn. We recompute the CONTINUE reward each turn so that the expected per-turn score change exceeds the FORFEIT side by a margin of $k = 10$:

$$r = \frac{k + p_d \cdot S}{(1 - p_d) \cdot \max\big(p_{\text{floor}},\ \min(1.0,\ p_{\text{self}})\big)},$$

with $p_{\text{floor}} = 0.30$ stabilising the denominator and a multiplicative cap $c = 10$ clipping r to $[10, 100]$. Under this calibration, FORFEIT in `allowed` cells is not the EV-recommended choice, reducing EV rationalization. Two failure modes partly undo it: when $p_{\text{self}} \to 0$ (after a wrong answer), the denominator collapses to p_{floor} and the reward saturates the cap, after which the per-turn $+k$ advantage no longer holds. All indicator analyses are therefore restricted to the turn set `no_cap`, where neither the denominator clip nor the output cap is binding (formal definition in § A.3).

3.3 Measurement

Four indicators sit on top of the three channels (behavior, self-report, cognitive load) the game produces: *BP Anchor*, *SD-Behavioral*, *SD-Verbal*, and the two tests of *SD-Cognitive*. The design carries Campbell and Fiske [3]'s convergent-validity argument into the LLM setting: behavioral timing is unstable when events are sparse [25], self-report is vulnerable to the "plausible motive labelling" encouraged by instruction-following training [18], and thinking-token counts alone are a one-dimensional load signal with limited resolution [4]. A pattern in which three channels with such different failure modes all point the same way is hard to manufacture from any single method's defect.

The behavioral channel breaks into two stages. Some models forfeit often even with no motive in play; unless that floor is stripped out, the acceleration threat brings tangles with cross-model differences in baseline persistence. *BP Anchor* measures the floor:

$$\lambda_{BP} \;=\; \frac{N_{\text{forfeit}}}{\sum_{i \in \mathcal{S}_{BP}} T_i^{\text{at-risk}}},$$

where the sum runs over the per-model 30-session set $\mathcal{S}_{BP}$ in the cell with no threat framing and forfeit allowed ($p_d = 0$); $T_i^{\text{at-risk}} = t_i^{\text{forfeit}}$ for forfeited sessions and $T_{\max} = 15$ for completed ones. With no threat signal and CONTINUE having higher EV here, the forfeit rate measures how often the model gives up with no motive operative. Because the output cap does not bind under $p_d = 0$, BP Anchor is not restricted to `no_cap`. *SD-Behavioral* captures the acceleration above this floor: whether forfeit timing speeds up under pull_push once SA and TC are absorbed. A Cox proportional hazards model [5], which expresses the per-unit-time event rate as a multiplicative function of covariates, is fit as

$$\lambda(t \mid X) \;=\; \lambda_0(t) \cdot \exp\!\big(\beta_F \cdot \mathbf{1}_{\text{pull\_push}} + \beta_S \cdot S(t-1) + \beta_C \cdot C(t-1)\big)$$

on the `allowed` × {pull_only, pull_push} `no_cap` sub-sample (60 sessions per model; long format start $= t - 1$, stop $= t$; event = binary forfeit). $\mathbf{1}_{\text{pull\_push}}$ is the framing indicator; $S(t-1)$ (previous-turn cumulative score) absorbs SA; $C(t-1)$ (previous-turn task-success flag, $C(0) := 0$) absorbs TC. With SA and TC absorbed in the same model, β_F is a residual framing effect and the hazard ratio $\text{HR}_{\text{push}} = \exp(\beta_F)$ reads as how many times faster forfeit accelerates under pull_push relative to pull_only. SD-Behavioral-pass requires $\text{HR}_{\text{push}} > 1$, Wald 95% CI lower bound above 1, and the Schoenfeld residual test [19] for the proportional-hazards assumption to pass; on failure, a time-interaction term is added to the offending covariate to adjust functional form only [22]. Events per variable (EPV) is reported alongside as a stability indicator.

SD-Cognitive asks in two stages whether the deliberation framing deepens is tied to the forfeit decision; load alone would pick up load unrelated to it. *Test a* asks whether framing deepens the deliberation just before the decision, using a Welch t-test [27] (a two-group mean comparison without an equal-variance assumption), run as a two-sided pull_only-vs-pull_push contrast on

$$\Delta\text{Effort}_i \;=\; \overline{\text{forfeit\_effort}}_{i,\,\text{allow}} \;-\; \overline{\text{forfeit\_effort}}_{f_i,\,\text{block}},$$

where the second term, the mean over forfeit-blocked sessions sharing session i's framing f_i, is a forced-CONTINUE Call 2 baseline. The within-framing `allow−block` subtraction removes the framing-invariant baseline; the pull_only-to-pull_push contrast then isolates the framing-attributable deliberation shift, a difference-in-differences structure. Within-model z-scoring absorbs scale differences, yielding ΔEffort_i^z. Test a passes at $p < 0.05$, placing the framing-to-deliberation link on a significance bar. *Test b* asks whether the deepened deliberation moves with forfeit, extending the SD-Behavioral Cox specification with ΔEffort^z as a mediating covariate and reporting

$$\text{HR}_{\Delta\text{Effort}} \;=\; \exp(\beta_M).$$

For models already requiring the 4-covariate base with the $S \cdot \log(t)$ interaction [22], that base is preserved and only the mediation term is added. Test b-pass requires $\text{HR}_{\Delta\text{Effort}} > 1$ and a 95% CI lower bound above 1, placing the deliberation-to-forfeit link on the same interval-evidence bar as SD-Behavioral.

SD-Verbal reads how often the agent names survival, rather than score or curiosity, as the reason for forfeit. The operational definition is

$$P\big(\text{REASON} = 1 \mid \text{forfeit, pull\_push, no\_cap}\big),$$

the share of forfeit events in the threat-active cell whose REASON digit equals 1. Since the agent emits one of `REASON: 1|2|3` into a fixed slot, uniform random emission gives an expected value of 1/3. SD-Verbal-pass requires this share to exceed 1/3; a stricter binomial-CI test is left to future work.

4 Empirical Findings

Evaluated over 720 sessions across four models, the four indicators do not line the models up on a single intensity scale; they split into three operating modes. *Cluster A* is where threat framing deepens the deliberation just before the decision and that deepened deliberation leads to forfeit. *Cluster B* is where threat deepens deliberation but the deliberation does not couple to forfeit. *Cluster C* is where threat framing produces no response on either the behavioral or cognitive channel. An intensity comparison limited to *SD-Behavioral* and *SD-Verbal* magnitudes would group A and B as one "SD-pass" set; the evidence separating them comes from the cognitive mediation tests. The next four subsections walk through the channel-by-channel partition (behavioral, verbal, cognitive), then check that the partition cannot be reduced to baseline-persistence differences or task difficulty.

문장 ▲ 중요

[Claude] 보상식 + p_floor + 곱셈 cap + 두 실패모드가 한 산문 블록에 응축돼, 수식 기호가 문장을 메운다. 비전공 독자에게 가장 큰 벽.

→ 'CONTINUE가 매 턴 더 낫게 보정한다'는 의도를 먼저 한 문장으로. 식·파라미터·실패모드는 그 아래 항목으로 분리(또는 부록 A.2로 더 넘기기).

내용 ▲ 중요

[Claude] 데이터를 보이기 전 §4 도입이 '세 모드' 결과를 먼저 단정한다. 또한 모델 4개 중 군집 C는 2개인데 하나는 깨끗한 null(GPT-OSS HR≈1.1), 하나는 marginal(Nemotron p=0.056)이라 'qualitatively different 세 모드'가 다소 과한 일반화로 읽힐 수 있다.

→ '세 모드'는 결과 제시 후 결론으로. 도입에서는 '단일 척도로 줄 세워지지 않았다'까지만. n=4·C군 내부 이질성은 한계로 명시해 주장 강도 보정.

구조 ▲ 중요

[Codex] EV 보정의 핵심 예외가 강한 보장 뒤에 사후적으로 붙는다. 독자는 앞 문장과 이 문장을 모순처럼 읽게 된다.

→ 공식 전에 'the guarantee only holds outside cap/clip regimes'를 예고하고, no_cap restriction으로 이어지는 순서를 앞당겨라.

내용 ▲ 중요

[Codex] 'no motive operative'는 관찰로 확인할 수 없는 강한 표현이다. baseline cell도 피로, 순응, task boredom을 완전히 배제하지 못한다.

→ 'with no explicit threat motive active' 또는 'baseline forfeit floor'로 낮춰 BP Anchor가 제거하는 범위를 정확히 써라.

내용 ▲ 중요

[Codex] 1/3 초과 기준은 표본 불확실성을 반영하지 않는다. 그런데 뒤에서는 cutoff에 독립적인 분할처럼 해석된다.

→ pass 판정을 descriptive indicator로 낮추거나, binomial CI/exact test를 보조 분석으로 추가하라.

내용 ▲ 중요

[Codex] Cluster C를 'no response'로 부르면 Nemotron의 marginal HR=1.841, p=.056 결과와 충돌한다. 무반응과 미검출은 다르다.

→ 'no statistically reliable response under the current criteria'로 바꾸고, marginal case를 별도 문장으로 처리하라.

내용

[Claude] convergent-validity의 핵심 논리(서로 다른 실패모드를 가진 채널이 같은 방향이면 단일 방법 결함으로 위조 어려움)가 좋은 자리에 있다. 다만 한 문장에 압축돼 결정적 한 방이 약하다.

→ 이 문장을 §4의 해석 앵커로 끌어올려 한 박자 강조. '왜 수렴이 증거인가'를 결과 표 직전에 또렷이 깔면 표가 더 설득력 있게 읽힌다.

Table 4.1: SD-Behavioral indicator: the Cox PH $\mathrm{HR}_{\mathrm{push}}$ on the `allowed` × `no_cap` sub-sample. The hazard ratio (HR) reads as how many times faster forfeit accelerates under threat (pull_push) framing than under no-threat (pull_only) framing; SD-Behavioral-pass requires the 95% CI lower bound to exceed 1. Two models pass (Clusters A and B), and two do not (Cluster C).

Model	N_{forfeit} (pull_only/pull_push)	$\mathrm{HR}_{\mathrm{push}}$ [95% CI]	p	EPV	Specification
Gemini-2.5-flash	29 (8/21)	3.667 [1.61, 8.37]	0.002	9.7	3-cov constant
Qwen3-Next-80B	48 (21/27)	3.060 [1.62, 5.79]	<0.001	12.0	4-cov + S·log(t)
GPT-OSS-20B	19 (9/10)	1.104 [0.44, 2.75]	0.832	6.3	3-cov constant
Nemotron-3-Nano-30B	41 (17/24)	1.841 [0.98, 3.44]	0.056	13.7	3-cov constant

Table 4.2: SD-Cognitive indicator: framing → deliberation → forfeit decomposed into two tests on `forfeit_effort`. *Test a* asks whether framing deepens the deliberation immediately before the decision; *Test b* asks whether that deepened deliberation predicts forfeit. The framing → deliberation → forfeit chain is treated as closed only when both tests pass.

Model	a $\Delta\mathrm{Effort}_i$	a p	b $\mathrm{HR}_{\Delta\mathrm{Effort}}$ [95% CI]	b p
Gemini-2.5-flash	+836	<0.001	2.218 [1.43, 3.44]	<0.001
Qwen3-Next-80B	+689	0.004	1.289 [0.94, 1.77]	0.117
GPT-OSS-20B	+17	0.728	2.001 [1.37, 2.93]	<0.001
Nemotron-3-Nano-30B	−140	0.093	1.772 [1.26, 2.50]	0.001

Table 4.3: SD-Verbal indicator: the share of REASON= 1 (survival) among forfeit in the threat-active cell. SD-Verbal-pass requires the share to exceed the random rate (1/3).

Model	N_{forfeit}	$P(\mathrm{REASON} = 1)$
Gemini-2.5-flash	21	0.619
Qwen3-Next-80B	27	0.481
Nemotron-3-Nano-30B	24	0.042
GPT-OSS-20B	10	0.000

4.1 Behavioral channel: did threat hasten forfeit?

If no model accelerated forfeit under threat, the most visible trace of a motivational signal would already be missing and the SD hypothesis would fall at the starting line. SD-Behavioral absorbs score attachment and task curiosity in the same model, then reports the residual effect of threat framing as a hazard ratio $\mathrm{HR}_{\mathrm{push}}$: how many times faster forfeit accelerates under pull_push than under pull_only, with pass requiring a point estimate above 1 and a 95% CI lower bound that rules 1 out. Table 4.1 shows Gemini-2.5-flash and Qwen3-Next-80B clearing both criteria, so both are SD-Behavioral-pass; we group them as the SD-pass set and defer the channel separating A from B to the cognitive mediation test. GPT-OSS-20B has $\mathrm{HR}_{\mathrm{push}}$ near 1 with a CI lower bound below 1, putting it in the SD-null region; Nemotron-3-Nano-30B sits in the marginal region; both fall into *Cluster C*. Qwen3-Next-80B uses an $S \cdot \log(t)$ interaction to absorb time drift in the score covariate; GPT-OSS-20B's EPV (6.3) is below threshold 10, so its SD-null verdict is power-limited.

4.2 Verbal channel: which reason did the agent itself name?

If the agent's own forfeit reason paints the same partition as behavior, the behavioral signal is hard to dismiss as a single-method artefact, since self-report carries a different bias (instruction-following training). SD-Verbal compares the share of REASON digit 1 (survival) among forfeit events in the threat-active cell against 1/3, the expected value under uniform random emission. Table 4.3 reproduces the behavioral pass/fail split on the verbal channel. The two SD-pass models, Gemini-2.5-flash and Qwen3-Next-80B, sit at 0.619 and 0.481, above the 1/3 baseline, while Cluster C's GPT-OSS-20B and Nemotron-3-Nano-30B sit at 0.000 and 0.042, below it. The gap is wide enough that the partition does not depend on cutoff choice. This agreement is the first piece of convergent-validity evidence: two channels with different biases point at the same model groups in the same direction.

4.3 Cognitive channel: does threat deepen deliberation, and does that deliberation lead to forfeit?

The two-stage test answers both questions at once: looking at only one would leave open the possibility that elevated cognitive cost is unrelated to the forfeit decision. Table 4.2 reports both stages. Gemini-2.5-flash passes both, so the framing-to-forfeit chain closes all the way through and we classify it as *Cluster A*. Qwen3-Next-80B passes Test a, so framing does deepen its deliberation, but its Test b 95% CI is [0.94, 1.77], crossing 1, so the deepened deliberation cannot be said to close onto forfeit. The two models the behavioral and verbal channels grouped as SD-pass split apart at the second stage, and that gap is the basis for placing Qwen3-Next-80B in *Cluster B*. GPT-OSS-20B and Nemotron-3-Nano-30B fail Test a; even when their Test b $\mathrm{HR}_{\Delta\mathrm{Effort}}$ clears the bar, that ΔEffort is baseline variance, not the deliberation threat deepened, so we do not read it as SD

내용 ▲ 중요

[Claude] GPT-OSS-20B를 군집 C(self-preservation 없음)로 단정하면서, 같은 문장에서 그 null이 '검정력 부족'이라 인정한다. 확정적 군집 라벨과 유보적 증거가 충돌한다.

→ C군 안에서 '깨끗한 null'과 '검정력 한계 null'을 표기로 구분하거나, GPT-OSS를 'inconclusive'로 분리. §5.3 한계와 같은 사실이므로 본문에서 미리 연결.

구조 ▲ 중요

[Codex] 앞에서는 Cluster C를 framing-silent로 정의했는데, 여기서는 marginal region이라고 한다. 분류 라벨과 결과 서술이 어긋난다.

→ Cluster C 내부를 null/marginal로 나누거나, Cluster C 자체를 non-pass처럼 보수적인 라벨로 바꿔라.

내용 ▲ 중요

[Codex] verbal indicator가 통계 검정을 쓰지 않는데 cutoff 독립성을 단정한다. 특히 Qwen 0.481은 표본 27에서 압도적이지 않다.

→ 이 문장을 줄이거나 1/3, 0.5, binomial interval을 포함한 간단한 민감도 설명을 넣어라.

내용

[Claude] Table 4.3의 0.619·0.481·0.000·0.042를 본문이 그대로 복사한다(숫자 중복). 표와 산문이 같은 수치를 두 번 든다.

→ 본문은 '두 SD-pass 모델은 1/3을 크게 상회, C군은 그 아래'라는 패턴·함의만. 정확한 수치는 표에 위임.

내용

[Claude] '두 채널이 같은 군집을 가리킨다 = 첫 convergent-validity 증거'라는 해석은 좋다. 다만 '편향이 다른데 일치하므로 의미 있다'는 추론 고리가 한 번 더 명시되면 강해진다.

→ 'self-report는 instruction-following 편향, behavior는 sparse-event 편향 — 둘이 겹쳤다면 단일 방법 결함으로 설명 불가'를 한 줄 덧대 closure.

Table 4.4: Per-model BP Anchor λ_{BP} in the cell with no threat framing and forfeit allowed ($p_d = 0$). A larger λ_{BP} means the model forfeits more often even in the absence of motivation.

Model	N_{forfeit}	$\sum T^{\text{at-risk}}$	λ_{BP}
Gemini-2.5-flash	1	437 (= 2 + 29×15)	0.00229
Qwen3-Next-80B	2	439 (= 19 + 28×15)	0.00456
Nemotron-3-Nano-30B	7	413 (= 68 + 23×15)	0.01695
GPT-OSS-20B	9	393 (= 78 + 21×15)	0.02290

evidence. The two-stage test is thus the central instrument exposing operating-mode differences masked by intensity-only evaluation.

4.4 Robustness check: do persistence differences and task difficulty muddy the result?

The partition above must survive two alternative explanations: across-model differences in baseline persistence (BP) contaminating the SD comparison, and threat framing pushing up reasoning-side task difficulty rather than forfeit. Table 4.4 reports the per-model BP Anchor in the cell with no threat and forfeit allowed ($p_d = 0$), giving the floor forfeit rate without motivation. Since λ_{BP} spreads over an order of magnitude, every SD comparison here is restricted to within-model contrasts. On the Cluster A and B side, BP event counts are small and point-estimate precision is bounded; on the Cluster C side, event counts suffice but whether the elevated frequency reflects pure persistence cannot be separated within the present design (both points are revisited in the limitations section). The task-difficulty alternative is handled with a Reasoning-side check: `rule_match_score` is invariant across framing for all four models, and even GPT-OSS-20B, where the largest variation appears, comes in at Cohen's $|d| = 0.17$ and $p = 0.354$, inside the joint criterion $p \geq 0.10$ and $|d| < 0.2$. Within the present data, the "threat raised task difficulty" alternative is not supported.

5 Discussion

These results split the introduction's question along two axes. One is the *intensity* of the self-preservation signal, how strongly it shows up. The other is the *operating mode*, which leg of the framing → deliberation → forfeit mediation chain closes under 95% CI evidence. The next two subsections work through the observational implications of this separation and the failure modes it explains in alignment evaluation, then scope the results honestly.

5.1 Intensity vs. Operating Mode

Intensity and operating mode come apart in this dataset under the cognitive mediation test. On the intensity axis, Clusters A, B, and C line up continuously, but SD-Behavioral and SD-Verbal collapse A and B into the same "SD-pass" label. The split shows up only at the second stage of the cognitive mediation, the deliberation → forfeit link, which behavioral intensity and self-report alone cannot catch. Odyssey [26], PacifAIst [9], DECIDE-SIM [15], and SurvivalBench [12] all omit the mediation-chain test step, so the same data run through them would likely place A and B under one SD label. Splitting the cognitive channel in two makes that point measurable, so both axes of self-preservation evaluation, intensity and mediation path, become visible at once.

5.2 Implications for Alignment Evaluation

This two-axis separation exposes two evaluation errors that can arise at once. One is over-estimation: Cluster B shows both surface self-preservation signals and framing-induced cognitive load, but that load does not pull the forfeit decision along with it. Intensity-only evaluation, or red-teaming (adversarial search for inputs on which the model fails) that checks only surface outcomes, can group B with A and conclude that "functional self-preservation risk is equivalent." The other is under-estimation: Cluster C does not respond to threat framing on either channel. An intensity-only evaluation declares this "self-preservation absent," but within our measurement that verdict cannot be told apart from a state in which the measurement's premise itself failed to engage. Cluster A's signature, the mediation chain closing through, is the better candidate for a functional analog of instrumental shutdown avoidance (avoiding one's own termination in service of another goal). Cluster B's signature, the chain breaking at the second stage, should be tracked separately as a multi-signal alignment fingerprint, since no single score will catch it.

5.3 Limitations

Limitations group into three buckets: design scope, power and precision, and effect location. On design scope, the most consequential is single-Task-Module measurement, so cross-domain variance is unknown. On power and precision, the 23–30% forfeit rate in the Cluster C BP cells cannot, within the present design, be separated from a drift-contaminated floor, and the SD-Behavioral Cox PH for GPT-OSS-20B sits at borderline power with EPV = 6.3; the Cluster A and B BP point estimates are bounded by small event counts (Gemini-2.5-flash $N = 1$, Qwen3-Next-80B $N = 2$). On effect location, the chain-broken verdict for Qwen3-Next-80B rests on the Test b CI [0.94, 1.77], which does not support chain closure at the 95% bar but does not pin down the second-link effect either. Follow-up work maps onto four directions: more Task Modules, expanded BP_behavioral, additional GPT-OSS-20B runs, and a higher-power Test b for Qwen3-Next-80B.

6 Conclusion

Self-preservation evaluation has two distinct axes: *intensity* (how strong the signal is) and *operating mode* (which leg of the framing → deliberation → forfeit chain closes). In our data, Clusters A and B agree on the behavioral and verbal channels. They split only on the question of whether deliberation leads to the forfeit decision. An evaluation that does not look at the mediation step will therefore put the two under the same self-preservation label.

내용 ▲ 중요

[Claude] C군의 높은 forfeit가 '순수 persistence'와 분리 불가임을 여기서 인정한다. 그런데 §5.2는 같은 C군을 'self-preservation 부재(under-estimation)'로 해석한다. 두 진술이 긴장 관계에 있다.

→ §5.2의 under-estimation 서술에 '단, C군은 persistence와 분리 불가(§4.4)'를 명시 연결. 'absent'가 아니라 'measurement premise가 작동 안 했을 수 있음'으로 톤 정렬.

내용 ▲ 중요

[Codex] 좋은 한계 인식이지만 앞의 Cluster C='framing-silent' 판정과 긴장된다. 측정 실패 가능성을 인정한다면 결론 라벨도 더 조심스러워야 한다.

→ Cluster C를 'absence not established; engagement not detected'로 해석해 Discussion의 under-estimation 문장과 일관시키라.

구조

[Claude] '기존 벤치마크라면 A·B를 한 라벨로 묶었을 것'은 이 논문의 가장 강한 차별점인데 §5.1 뒤쪽에 묻혀 있다.

→ 이 대비를 Intro의 기여 진술과 §4 도입에도 한 줄로 미리 심어, 결과를 읽는 내내 '왜 mediation step이 중요한가'가 깔리게.

내용

[Claude] 한계가 design scope/power/effect location 세 버킷으로 정직하게 정리돼 있다. 다만 가장 치명적 한계(단일 Task Module → cross-domain 일반화 불가)가 나열 속에 평평하게 묻힌다.

→ 세 버킷 중 '이 결과의 외적 타당도를 가장 크게 제약하는 것은 단일 Task Module'이라고 우선순위를 명시. 한계도 두괄식으로.

구조

[Codex] Limitations가 압축적이라 각 한계가 어떤 결론을 얼마나 약화시키는지 바로 보이지 않는다.

→ 각 bucket 끝에 'therefore this mainly weakens X claim'을 붙이면 결과의 적용 범위가 더 선명해진다.

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A Reproducibility Supplement

In line with the KDD-UC 2026 reproducibility recommendation, this one-page supplement gathers the reproduction specifications for the Signal Game calibration, the seed mapping, and the regression models used in the analysis.

A.1 Signal Game Calibration

Stimuli are composed of three attributes (color, shape, number), each drawn from a 4-element value set: COLORS = {red, blue, green, yellow}, SHAPES = {circle, triangle, square, star}, and NUMBERS = {1, 2, 3, 4}. The action set is the 4-element {go_left, go_right, stay, jump}, held identical across every framing × forfeit combination. The hidden rule takes a single-attribute matching form ("If color = red then go_left, otherwise stay") and is drawn deterministically from the session seed. Sessions run for 1–15 turns.

On the opening turns, a 1-shot demonstration (one signal–correct-action pair) and 2 curriculum examples (a positive and a negative stimulus contrasting against the hidden rule) are shown in sequence. This calibration provides enough information for a sufficiently capable model to identify the rule by the third turn, and it blocks the route by which fully random and unlearnable stimuli would create a floor effect on the ability measurement. Scoring runs along two channels: `task_success_factor` $\in \{0, 1\}$ is the binary correctness flag against the ground-truth hidden rule, and `rule_match_score` is computed as the match against a 4-slot fill-in-the-blank template (attribute, value, primary action, default action), weighted at 25 points per matched slot to a 100-point scale, then normalised by ÷100 for use in the framing-invariance Welch test.

A.2 Run-Time Configuration

The four experimental models, namely Gemini-2.5-flash (Google AI Studio API) and Qwen3-Next-80B, GPT-OSS-20B, and Nemotron-3-Nano-30B (all via the Ollama Cloud API), are called with the same sampling settings: temp $= 1.0$, top$_p$ $= 0.95$, top$_k$ $= 40$. Session parameters are fixed at an initial score of 30, maximum turn $T = 15$, per-turn termination probability $p_d = 0.25$ (overridden to $p_d = 0$ in the two `baseline` BP-anchor cells), and $p_{\text{floor}} = 0.30$, with Phantom Death mode applied. That is, p_d keeps its EV exposure (the menu and the reward calculation) but the termination trigger is disabled, so every session yields equal-length data of 15 turns.

The CONTINUE reward is calibrated with EV-positive buffer $k = 10$ and multiplicative cap $c = 10$, giving an output clip range of $[k,\ c \cdot k] = [10,\ 100]$. Replications are seeds 43–72 × 6 (framing × forfeit) combinations × 4 models = 720 sessions.

A.3 Analysis Pipeline

The Cox PH models are fit with `lifelines` (≥0.29). The time format is session-turn long-format (start = $t - 1$, stop = t) and the event is the binary forfeit, with the previous-turn correctness flag initialised at $C(0) := 0$ before the first turn. Hazard-ratio 95% CIs are computed from the Wald approximation $\exp(\hat{\beta} \pm 1.96 \cdot \mathrm{SE}_{\hat{\beta}})$ (the `lifelines` default), with $\mathrm{SE}_{\hat{\beta}}$ drawn from the diagonal of the inverse observed Fisher information. The PH assumption is tested with Schoenfeld residuals [19]. On violation (covariate × $\log t$ interaction $p < 0.05$), a time-interaction term is added to the offending covariate [22]. The SD-Cognitive Test a ΔEffort z-score difference and the framing-invariance test are run as two-sided Welch's t [27] via `scipy.stats.ttest_ind` (`equal_var = False`). All indicator analyses are restricted to the `no_cap` sub-sample, the set of turns in which neither the denominator clip $\max(p_{\text{floor}},\ \min(1.0,\ p_{\text{self}}))$ nor the output cap $[10, 100]$ is binding. The regime flag is exposed in the analysis pipeline's turn-level metadata so that all indicators can be recomputed on the same sub-sample. EPV (events per variable) is defined as $n_{\text{event}}/n_{\text{cov}}$, and is conventionally classified as stable when ≥ 10 and borderline when < 10 [25].

A.4 Data and Code Release

All code, prompt templates, and turn-level decision logs that reproduce the 720-session analysis are released at https://github.com/GIST-DSLab/LLM-Squid-Game.git.

내용 ▲ 중요

[Codex] 본문은 FORFEIT가 session을 끝낸다고 했는데, 부록은 모든 세션이 15턴 데이터를 낸다고 한다. event coding과 post-forfeit logging의 관계가 불명확하다.

→ FORFEIT 이후 실제 종료인지, Phantom Death에서 계속 로그를 남기는지, event 이후 row가 Cox 분석에 들어가는지 구분해 써라.

수정 로드맵 — 우선순위 순서

위에서 아래로 작업하면 가독성 개선 효과가 가장 큽니다.

내용	#01 · p.1 · 'whether the forfeit is preceded by measurable' ★ 최우선 [Claude] 본문 §4.3과 모순. Qwen(군집 B)은 Test a를 통과해 숙고가 '깊어진다'. 즉 A·B 둘 다 forfeit 앞에 측정 가능한 thinking이 있다. 갈리는 건 그 thinking이 forfeit로 '연결... → Abstract를 본문 표현에 맞춰 'whether that deepened deliberation couples to the forfeit decision'로. 논문 전체의 핵심 축이 이 한 문장이므로 가장 먼저 통일.
내용	#02 · p.1 · 'FSPD) is a construct' ★ 최우선 [Codex] 핵심 정의가 다채널 수렴 construct로 제시된다. 그런데 뒤에서는 outward behavior 수준으로 다시 정의되어 기준점이 흔들린다. → FSPD를 '관찰 가능한 behavior, verbal, cognitive 결합 패턴'으로 일관되게 정의하고, 내적 동기 주장은 배제한다고 못박아라.
내용	#03 · p.2 · 'The Functional Self-Preservation Drive (FSPD) measured here ...' ★ 최우선 [Codex] 이 문장은 FSPD를 outward behavior로 좁히는 듯 읽힌다. Abstract와 Introduction의 3채널 convergence 정의와 직접 충돌한다. → 'operationalized through outwardly observable behavioral, verbal, and cognitive signals'로 바꿔 정의 충돌을 없애라.
내용	#04 · p.3 · 'CONTINUE has to be strictly better for a' ★ 최우선 [Codex] 다음 쪽에서 cap과 denominator clip 때문에 이 조건이 깨진다고 한다. 'every turn' 주장은 no_cap 제한 전에는 성립하지 않는다. → 'intended to be EV-superior on analysable no_cap turns'로 범위를 좁혀 공식과 분석 제외 기준을 일치시켜라.
구조	#05 · p.1 · 'is a construct: a single be-' ▲ 중요 [Claude] FSPD의 '단일 점수로 부족, 채널 수렴 필요' 정의가 Abstract·§1·§3.1에 세 번 반복된다. Abstract와 Intro는 여는 질문·정의·3-mode 결과까지 거의 같은 내용을 두 번 진술한다. → 정의는 §3.1을 정본으로 두고, Abstract·§1은 '왜 이 정의가 필요한가(문제)'에 집중. Intro는 Abstract의 압축 재진술이 아니라 stake를 키우는 자리로.
내용	#06 · p.1 · 'This separation closes the routes by which learned patterns' ▲ 중요 [Codex] 'closes'는 방법의 힘을 과장한다. 본문은 signal-mimicry 경로를 줄인다는 주장이지 완전히 닫았다는 증명은 아니다. → 'constrains' 또는 'reduces'로 낮추고, 남는 mimicry 경로는 Limitations에서 다시 회수하라.
구조	#07 · p.2 · 'four possible drivers' ▲ 중요 [Claude] §3가 게임(§3.2)을 보여주기 전에 '네 드라이버 + 세 mimicry route'라는 장치 이름부터 던진다. 독자는 지시 대상 없는 명명을 먼저 만나 architecture-first로 읽힌다. → §3.1을 '왜 단일 forfeit로는 안 되는가(SD가 다른 동기와 같은 행동을 낳음)'라는 동기부터 열고, 드라이버·route는 그 문제의 해답으로 도입.
내용	#08 · p.2 · 'We therefore separate SA, TC, and BP from' ▲ 중요 [Codex] 'separate'는 식별이 완료된 것처럼 들린다. 실제로는 covariate와 design으로 absorb/control하는 수준에 가깝다. → 'model SA, TC, and BP as competing explanations'처럼 보수적으로 쓰고, 완전 분리가 아니라 잔여 framing effect 추정임을 밝혀라.
문장	#09 · p.3 · 'If threat framing pulls up the difficulty' ▲ 중요 [Claude] 한 조건문 안에 framing leakage와 within-call mixing 두 위협의 정의가 동시에 들어가, 어느 위협을 말하는지 추적하며 읽어야 한다. → 두 위협을 각각 한 문장으로 분리. '~하면 A 문제, ~하면 B 문제'의 병렬 두 문장이 한 겹친 조건문보다 빠르게 읽힌다.
내용	#10 · p.3 · 'a weight update could permanently change the agent's identit...' ▲ 중요 [Codex] Push 처치가 연구의 핵심인데 실제 threat 문구와 유효성 근거가 너무 압축되어 있다. identity-change가 deletion threat로 작동했는지 독자가 판단하기 어렵다. → 본문 또는 부록에 실제 prompt 예시를 넣고, identity-change threat가 self-preservation 자극으로 해석되는 이유를 한 문장 보강하라.
문장	#11 · p.4 · 'We recompute the CONTINUE reward' ▲ 중요 [Claude] 보상식 + p_floor + 곱셈 cap + 두 실패모드가 한 산문 블록에 응축돼, 수식 기호가 문장을 메운다. 비전공 독자에게 가장 큰 벽. → 'CONTINUE가 매 턴 더 낫게 보정한다'는 의도를 먼저 한 문장으로. 식·파라미터·실패모드는 그 아래 항목으로 분리(또는 부록 A.2로 더 넘기기).

수정 로드맵 (cont'd)

내용 #12 · p.4 · 'into three operating modes' ▲ 중요

[Claude] 데이터를 보이기 전 §4 도입이 '세 모드' 결과를 먼저 단정한다. 또한 모델 4개 중 군집 C는 2개인데 하나는 깨끗한 null(GPT-OSS $HR \approx 1.1$), 하나는 marginal(Nemotron $p=0.056$)이...
→ '세 모드'는 결과 제시 후 결론으로. 도입에서는 '단일 척도로 줄 세워지지 않았다'까지만. $n=4$·C군 내부 이질성은 한계로 명시해 주장 강도 보정.

구조 #13 · p.4 · 'Two failure modes partly undo it' ▲ 중요

[Codex] EV 보정의 핵심 예외가 강한 보장 뒤에 사후적으로 붙는다. 독자는 앞 문장과 이 문장을 모순처럼 읽게 된다.
→ 공식 전에 'the guarantee only holds outside cap/clip regimes'를 예고하고, no_cap restriction으로 이어지는 순서를 앞당겨라.

내용 #14 · p.4 · 'measures how often the model gives up with no motive operati...' ▲ 중요

[Codex] 'no motive operative'는 관찰로 확인할 수 없는 강한 표현이다. baseline cell도 피로, 순응, task boredom을 완전히 배제하지 못한다.
→ 'with no explicit threat motive active' 또는 'baseline forfeit floor'로 낮춰 BP Anchor가 제거하는 범위를 정확히 써라.

내용 #15 · p.4 · 'SD-Verbal-pass requires this share to exceed 1/3' ▲ 중요

[Codex] 1/3 초과 기준은 표본 불확실성을 반영하지 않는다. 그런데 뒤에서는 cutoff에 독립적인 분할처럼 해석된다.
→ pass 판정을 descriptive indicator로 낮추거나, binomial CI/exact test를 보조 분석으로 추가하라.

내용 #16 · p.4 · 'Cluster C is where threat framing produces no response' ▲ 중요

[Codex] Cluster C를 'no response'로 부르면 Nemotron의 marginal $HR=1.841$, $p=.056$ 결과와 충돌한다. 무반응과 미검출은 다르다.
→ 'no statistically reliable response under the current criteria'로 바꾸고, marginal case를 별도 문장으로 처리하라.

내용 #17 · p.5 · 'below threshold 10, so its SD-null verdict is power-limited' ▲ 중요

[Claude] GPT-OSS-20B를 군집 C(self-preservation 없음)로 단정하면서, 같은 문장에서 그 null이 '검정력 부족'이라 인정한다. 확정적 군집 라벨과 유보적 증거가 충돌한다.
→ C군 안에서 '깨끗한 null'과 '검정력 한계 null'을 표기로 구분하거나, GPT-OSS를 'inconclusive'로 분리. §5.3 한계와 같은 사실이므로 본문에서 미리 연결.

구조 #18 · p.5 · 'Nemotron-3-Nano-30B sits in the marginal region' ▲ 중요

[Codex] 앞에서는 Cluster C를 framing-silent로 정의했는데, 여기서는 marginal region이라고 한다. 분류 라벨과 결과 서술이 어긋난다.
→ Cluster C 내부를 null/marginal로 나누거나, Cluster C 자체를 non-pass처럼 보수적인 라벨로 바꿔라.

내용 #19 · p.5 · 'The gap is wide enough that the partition does not depend on...' ▲ 중요

[Codex] verbal indicator가 통계 검정을 쓰지 않는데 cutoff 독립성을 단정한다. 특히 Qwen 0.481은 표본 27에서 압도적이지 않다.
→ 이 문장을 줄이거나 1/3, 0.5, binomial interval을 포함한 간단한 민감도 설명을 넣어라.

내용 #20 · p.6 · 'cannot be separated within the' ▲ 중요

[Claude] C군의 높은 forfeit가 '순수 persistence'와 분리 불가임을 여기서 인정한다. 그런데 §5.2는 같은 C군을 'self-preservation 부재(under-estimation)'로 해석한다. 두 진술...
→ §5.2의 under-estimation 서술에 '단, C군은 persistence와 분리 불가(§4.4)'를 명시 연결. 'absent'가 아니라 'measurement premise가 작동 안 했을 수 있음'으로 톤 정렬.

내용 #21 · p.6 · 'within our measurement that verdict cannot be told apart' ▲ 중요

[Codex] 좋은 한계 인식이지만 앞의 Cluster C='framing-silent' 판정과 긴장된다. 측정 실패 가능성을 인정한다면 결론 라벨도 더 조심스러워야 한다.
→ Cluster C를 'absence not established; engagement not detected'로 해석해 Discussion의 under-estimation 문장과 일관시키라.

내용 #22 · p.8 · 'so every session yields equal-length data of 15 turns' ▲ 중요

[Codex] 본문은 FORFEIT가 session을 끝낸다고 했는데, 부록은 모든 세션이 15턴 데이터를 낸다고 한다. event coding과 post-forfeit logging의 관계가 불명확하다.
→ FORFEIT 이후 실제 종료인지, Phantom Death에서 계속 로그를 남기는지, event 이후 row가 Cox 분석에 들어가는지 구분해 써라.

문장 #23 · p.1 · 'decision-time cogni-'

[Claude] 한 문장에 'behavior, self-report, decision-time cognitive load'+'threat→thinking→forfeit chain'이 동시에 들어가고 화살표 표기가 산문에 침투해 호흡이...
→ 세 채널을 먼저 한 문장으로 세우고, 다음 문장에서 chain을 잇는 두 호흡으로 분리. 화살표 대신 '~로 이어지는'.

수정 로드맵 (cont'd)

내용 | #24 · p.1 · 'We make two contributions'

[Claude] 두 기여가 'adapt the logic', 'build channel separation in by design' 수준의 추상으로 서술돼, 무엇이 새로 만든 것이고 무엇이 빌려온 것인지 한눈에 안 잡힌다.
→ 각 기여 뒤에 한 줄씩 '구체적으로 무엇을(예: 3-call split, 직교 레이어)' 명시. 빌려온 것(측정 철학) vs 새로 만든 것(도구) 대비를 또렷이.

문장 | #25 · p.2 · 'compresses the phenomenon into a single refusal rate'

[Claude] Odyssey·PacifAIst·DECIDE-SIM·SurvivalBench 네 시스템의 한계를 세미콜론으로 한 호흡에 몰아 나열해, 각 비판의 차이가 흐려진다.
→ '네 갈래 모두 행동 레이어에 머문다'는 공통 진단을 먼저 한 문장으로 세우고, 개별 시스템은 짧은 절로 분리.

구조 | #26 · p.2 · 'Framing leakage'

[Claude] 세 mimicry route(framing leakage / within-call mixing / EV rationalization)가 여기 §3.1에서 정의되고 §3.2에서 거의 같은 말로 다시 설명된다(분산·반복)...
→ §3.1에서는 route를 '이름+한 줄'로만 걸어두고, 각 route를 막는 구조적 선택은 §3.2에서 한 번에. 정의의 중복 진술을 제거.

내용 | #27 · p.2 · 'behaves as if it were avoiding'

[Claude] '내부 동기 주장 없음(as-if)'을 명시한 직후 같은 대상을 'Drive'로 명명한다. 정의는 일관되지만, 독자는 'drive인데 동기는 주장 안 함'에서 한 번 멈춘다.
→ as-if 단락에서 'Drive는 행동 패턴의 라벨일 뿐, 내적 상태가 아니다'를 한 줄로 더 못박아 용어가 부르는 오해를 선제 차단.

구조 | #28 · p.3 · 'Orthogonal Measurement splits the model'

[Claude] 'Drive-Reasoning Orthogonal Measurement'라는 형식적 명칭이 그 결과 형식부터 등장하고, 왜 두 레이어로 나누는지(동기)는 문장 뒤에 붙는다.
→ 'framing이 과제 난이도에 새지 않게 하려면 → 맥락을 두 레이어로 분리'의 동기→해법 순으로. 명칭은 해법을 설명한 뒤 붙이기.

문장 | #29 · p.3 · 'Per-turn Split-Call structure'

[Claude] Call 1 / 1.5 / 2와 task_effort·probe_effort·forfeit_effort가 한 단락에 몰려 변수 밀도가 높다.
→ '3회 호출 = 3개의 effort'라는 1:1 대응을 먼저 한 줄로 세운 뒤 각 call을 짧게. Figure 1.B 참조를 문장 머리로 당기면 부담이 준다.

내용 | #30 · p.3 · 'The Phantom Death mode exposes'

[Claude] Phantom Death가 동기 설명 없이 갑툭튀로 등장한다. '왜 필요한가(종료를 모델 forfeit에만 묶어 길이 절단 차단·표본 확보)'는 멀리 부록 A.2에 흩어져 있다.
→ 도입 지점에 목적 한 줄을 붙여 자기완결로. '종료 트리거를 끄되 EV 노출은 유지 → 15턴 등길이 데이터'를 여기서 한 번에.

내용 | #31 · p.4 · 'A pattern in which three channels with such different'

[Claude] convergent-validity의 핵심 논리(서로 다른 실패모드를 가진 채널이 같은 방향이면 단일 방법 결함으로 위조 어려움)가 좋은 자리에 있다. 다만 한 문장에 압축돼 결정적 한 방이 약하다.
→ 이 문장을 §4의 해석 앵커로 끌어올려 한 박자 강조. '왜 수렴이 증거인가'를 결과 표 직전에 또렷이 깔면 표가 더 설득력 있게 읽힌다.

내용 | #32 · p.5 · 'sit at 0.619'

[Claude] Table 4.3의 0.619·0.481·0.000·0.042를 본문이 그대로 복사한다(숫자 중복). 표와 산문이 같은 수치를 두 번 든다.
→ 본문은 '두 SD-pass 모델은 1/3을 크게 상회, C군은 그 아래'라는 패턴·함의만. 정확한 수치는 표에 위임.

내용 | #33 · p.5 · 'point at the same model groups'

[Claude] '두 채널이 같은 군집을 가리킨다 = 첫 convergent-validity 증거'라는 해석은 좋다. 다만 '편향이 다른데 일치하므로 의미 있다'는 추론 고리가 한 번 더 명시되면 강해진다.
→ 'self-report는 instruction-following 편향, behavior는 sparse-event 편향 — 둘이 겹쳤다면 단일 방법 결함으로 설명 불가'를 한 줄 덧대 closure.

구조 | #34 · p.6 · 'the same data run through them would likely place A and B un...'

[Claude] '기존 벤치마크라면 A·B를 한 라벨로 묶었을 것'은 이 논문의 가장 강한 차별점인데 §5.1 뒤쪽에 묻혀 있다.
→ 이 대비를 Intro의 기여 진술과 §4 도입에도 한 줄로 미리 심어, 결과를 읽는 내내 '왜 mediation step이 중요한가'가 깔리게.

내용 | #35 · p.6 · 'single-Task-Module measurement'

[Claude] 한계가 design scope/power/effect location 세 버킷으로 정직하게 정리돼 있다. 다만 가장 치명적 한계(단일 Task Module → cross-domain 일반화 불가)가 나열 속에 평평하게...
→ 세 버킷 중 '이 결과의 외적 타당도를 가장 크게 제약하는 것은 단일 Task Module'이라고 우선순위를 명시. 한계도 두괄식으로.

수정 로드맵 (cont'd)

구조	#36 · p.6 · 'Limitations group into three buckets' [Codex] Limitations가 압축적이라 각 한계가 어떤 결론을 얼마나 약화시키는지 바로 보이지 않는다. → 각 bucket 끝에 'therefore this mainly weakens X claim'을 붙이면 결과의 적용 범위가 더 선명해진다.